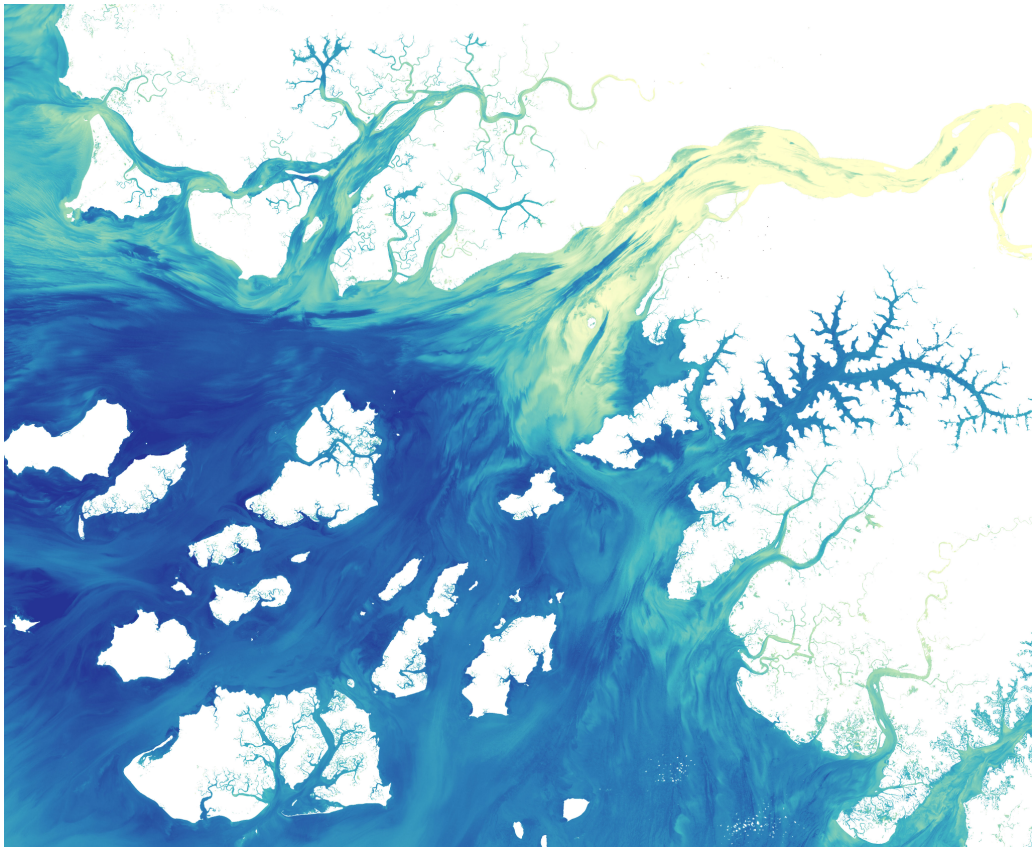


Sierra Vista 2022 • Division B

Dynamic Planet KEY

Section 1 (23 points)

1) What feature is shown at letter C? (1 pt)

a. Intermittent pond
b. Marsh
c. Submerged marsh
d. Spring
e. Mangrove forest

2) What type of feature is Janicke Slough (near the center)? (1 pt)

a. Oxbow lake
b. Yazoo stream
c. Pingo
d. Kettle lake
e. Esker

3) The stream length from A to B is 5.9 km. Estimate the sinuosity of the Snoqualmie River from point A to B. (2 pts)

The straight line distance is about 2.4 km (1 pt). Thus, S = 5.9 km / 2.4 km = 2.5. (1 pt) Accept answers from 2.3 to 2.7 if the work shown is reasonable.

4) This section of the Snoqualmie River can be best classified as: (1 pt)
a. Braided
b. Meandering
c. Straight

Tanner and Snoqualmie are two nearby towns with stream gauges along the Snoqualmie River. The hydrographs with predicted river level and discharge are given below. The red line indicates the time at which the hydrograph was produced.

[TB 1] 5) Using the hydrographs, which town is downstream relative to the other? Give two reasons for your answer. (4 pts)

Snoqualmie is downstream to Tanner (1 pt). We can see that the peak discharge of the precipitation event around Jan 7 is earlier for Tanner than Snoqualmie, suggesting that Tanner is upstream (1.5 pts). Also, the discharge at Snoqualmie is greater than Tanner, which is reasonable only if Tanner is upstream of Snoqualmie (1.5 pts).

6) At Snoqualmie, how much greater was the actual peak discharge than the median predicted peak discharge on January 7? Round to the nearest thousand cubic feet per second. (2 pts)

From the hydrograph, the actual discharge was about 14500 ft^3/s, while the median predicted discharge was 3500 ft^3/s. The difference is 11,000 ft^3/s!

How do hydrologists predict peak discharge and flood events? Probabilistic models using historical data can give the chance that a flood of a given intensity will occur in a given year.

7) A flood with an annual exceedance probability of 2% is also known as a: (1 pt)

a. 100-year flood
b. 50-year flood
c. 20-year flood
d. 2-year flood

8) What is the chance that a flood with an annual exceedance probability of 2% will occur at least once in 100 years? (3 pts)

Flood with AEP of 2% means that it has 2% of occurring in any given year. Thus the chance that it *won't* occur in 100 years is (0.98)^100 = 0.133 (1 pt). The chance that it *will* occur at least once is 1 - 0.133 = 0.867 → 86.7%. (2 pts)

Physical rainfall-runoff models can also forecast flooding. One of the simplest methods for estimating discharge is the Rational Formula, which assumes that the peak discharge of a flood is linearly proportional to the average rainfall intensity.

[TB 2] 9) Let's find the Rational Formula using dimensional analysis. If the rainfall intensity i is given in m/s and catchment area A is given in m^2, write an equation for the peak discharge Q. (*Hint: units should be consistent in the equation. You will need to multiply by a dimensionless constant C that represents the degree of runoff versus infiltration*) (3 pts)

Using dimensional analysis and the assumption that $Q \propto i$, we can reasonably write $Q = CiA$. This way, Q has units m^3/s. (3 pts)

Unfortunately, the Rational Formula isn't a very reliable way to estimate peak discharge. One reason is that the runoff constant C is not really constant.

10) What numerical range *must* C fall within? Explain. (2 pts)

C should be within 0 and 1 (1 pt). In an ideal basin where all precipitation goes to contribute to peak discharge, C would be 1. Physically, this is the maximum possible value of C. (1 pt for a reasonable explanation)

11) Consider the physical meaning of C (the degree of runoff versus infiltration) and its mathematical representation in the Rational Formula. For each set of conditions below, circle the one that would have a greater C. (1 pt each)

A. Sandy soil / **Clay soil**

B. Slope of 0.05 / **Slope of 0.10**

C. **Saturated soil** / Dry soil

Section 2 (29 points)

1) Which of the following rocks is the most permeable? (1 pt)

a. Granite
b. Mudstone
c. Slate
d. Sandstone
e. Vesicular basalt

2) Which of the following is closest to the total percentage of liquid freshwater located in aquifers? (1 pt)

a. 1.2%
b. 3.5%
c. 30.1%
d. 55%
e. 96%

The following true or false questions are 1 point each.

3) **True**/false: groundwater extraction can cause land subsidence.

4) **True**/false: marine transgression can lead to saline groundwater

5) True/**false**: a perched aquifer cannot form above the local water table

6) **True**/false: groundwater elevation lines generally follow topographic elevation lines.

Hydraulic head is a convenient way to represent the total energy of a fluid.

7) The gravitational potential energy U of some fluid of mass m at height h given by $U = mgh$, where g is the acceleration due to gravity. Find the gravitational potential energy for fluid at the top of the Piezometer 1. (2 pts)

$U = mg(\psi_1 + z)$**. (2 pts)**

8) Find the difference in gravitational potential energy between Piezometer 1 and Piezometer 2. Given this information and considering the figure, which way will groundwater flow? (3 pts)

$\Delta U = mg(\psi_2 - \psi_1)$ **or** $\Delta U = mg(\psi_1 - \psi_2)$ **(1 pt). Since** $\psi_2 > \psi_1$**, groundwater will flow towards the left (from 2 to 1). (2 pts)**

The pressure P exerted at the bottom of a stationary (hydrostatic) fluid is $P = \rho g h$ where ρ is the density of the fluid, g is the acceleration due to gravity, and h is the height of the fluid column.

9) Find the pressure head ψ in terms of the pressure (P) at the bottom of a piezometer. (2 pts)

We can see in the figure that ψ **is the height of the fluid column in the piezometer. Thus, rearranging the hydrostatic equation we get** $\Psi = \frac{P}{\rho g}$**. (2 pts)**

10) Considering the result in 9, is the pressure at the bottom of piezometer 1 (P_1) or the pressure at the bottom of piezometer 2 (P_2) greater? (1 pt)

$P_2 > P_1$ **because** $\psi_2 > \psi_1$**. (1 pt)**

Now, it's time to put it all together. Hydraulic head h represents the height that groundwater would reach if all of its energy (including gravitational potential and pressure) was converted into gravitational potential.

11) Use the equation you wrote in (7) to solve for height–this is the hydraulic head! Then substitute in the equation in (9) to get the final equation for the hydraulic head. (2 pts)

$h = \frac{U}{mg} = (\psi + z)$ **(1 pt). Subbing in** $\psi = \frac{P}{\rho g}$**, we get** $h = \frac{P}{\rho g} + z$**. (1 pt)**

[TB 3] 12) You might be wondering, what about kinetic energy? Let's consider a case where the groundwater is also moving at velocity v and has kinetic energy $\frac{1}{2}mv^2$. If all of the kinetic energy is converted to gravitational potential, what height h would the water column reach? (2 pts)

$\frac{1}{2}mv^2 = mgh$ **(1 pt). Mass cancels, and rearranging we have** $h = \frac{v^2}{2g}$**. (1 pt)**

13) If we didn't ignore kinetic energy, the height you found above would need to be added to the hydraulic head. Why is the kinetic energy term usually neglected for groundwater calculations involving hydraulic head? (2 pts)

Groundwater moves very slowly, so the kinetic energy of the flow is negligible. (2 pts)

14) Which of the following sets of minerals constitutes most rocks in areas of karst topography? (1 pt)

a. Quartz and feldspar
b. Calcite and dolomite
c. Muscovite and pyrite
d. Feldspar and biotite
e. Olivine and pyroxene

15) How does climate affect the maturity of karst landforms? (1 pt)

a. Warmer, wetter climates lead to stronger karstification
b. Cooler, drier climates lead to stronger karstification
c. The effect of climate on karstification largely depends on the local rock composition
d. Climate does not have a significant effect on the degree of karstification

16) Select all of the following regions that have extensive karst topography. (1 pt)

a. Washington
b. Southern China
c. Florida
d. Nepal

[TB 4] 17) Which of the following best describes a karstic aquifer? (1 pt)
a. Homogeneous and isotropic
b. Homogeneous and anisotropic
c. Non-homogeneous and isotropic
d. Non-homogeneous and anisotropic

18) Briefly explain how extraction of coastal groundwater may lead to saltwater intrusion. (2 pts)

A cone of depression (1 pt) forms at that well, which allows nearby salty water to flow towards the well (1 pt).

19) Refer to figure 5. Briefly explain why there is *saltwater* intrusion (top) at an ocean-aquifer interface rather than *freshwater* intrusion (bottom). (2 pts)

Saltwater is denser than freshwater, so it exerts greater hydrostatic pressure; it also has greater hydraulic head. (2 pts; either explanation is sufficient)

20) What is the name of the semi-empirical relationship that relates the depth of saltwater intrusion relative to the height of an unconfined freshwater aquifer above sea level? (1 pt)

Ghyben-Herzberg relation (1 pt).

Section 3 (23 pts)

1) Lake Agassiz is best classified as a ____ lake. (1 pt)

a. Tectonic
b. Pluvial
c. Proglacial
d. Volcanic
e. Rift

2) Match the following terms to their definitions. (0.5 pt for each)

a. Monomictic **ii**
b. Epilimnion **iv**
c. Pycnocline **iii**
d. Meromictic **i**

i. A lake that does not mix
ii. A lake that mixes once per year
iii. A sharp density gradient
iv. The surface layer of a stratified lake

Refer to Figure 6 and Figure 7 for the following _ questions.

3) Which hemisphere is Long Lake in? (1 pt)

Northern Hemisphere.

4) Briefly explain why the 15 deg C isotherm slopes down between mid-June and October. (2 pts)

The lake warms throughout the summer (2 pts); stratification prevents the warm water from penetrating very quickly downwards, so the isotherm slopes down slightly over time.

5) What happened in mid-October? Suggest a possible cause for this event. (2 pts)

Mixing/overturning (1 pt). Probably wind-driven (1 pt).

6) Is Long Lake a dimictic, monomictic, or amictic lake? Justify your answer. (2 pts)

Dimictic (1 pt). We see that stratification appears in the summer and is preceded and succeeded by two mixing events in spring and autumn, respectively (1 pt).

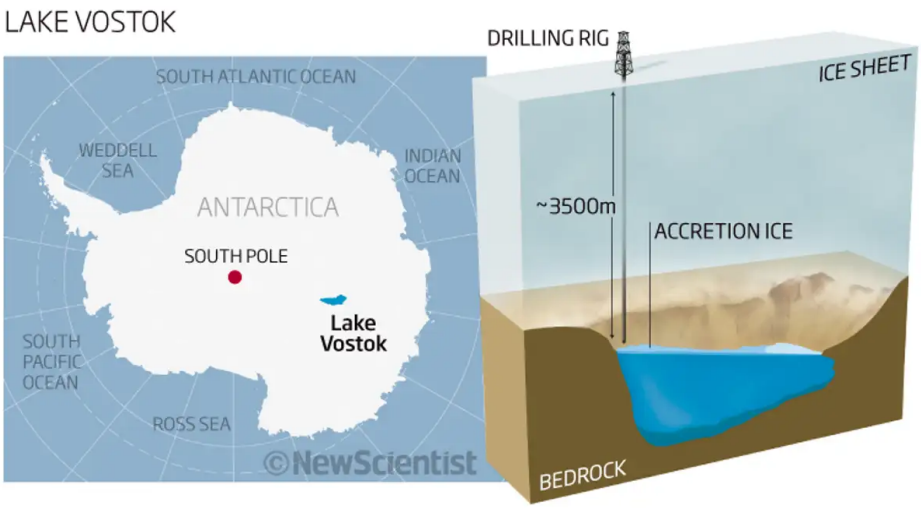
7) **True**/false: the presence of more chlorophyll in the water increases the light attenuation. (1 pt)

8) **True**/false: the total dissolved oxygen during June is greater than the total dissolved oxygen during August. (1 pt)

9) In what month did the total daily mean chlorophyll reach its maximum? Interpret what this means in terms of the lake's productivity. (2 pts)

July (1 pt). Chlorophyll content is a measure of the photosynthetic biomass, which means the lake was the most productive in July (1 pt).

Lake Vostok is a subglacial lake in Antarctica that is overlain by almost 4 kilometers of ice.



10) In order for there to be liquid water in otherwise subfreezing temperatures, there must be a heat source. Speculate a source of heat that maintains Lake Vostok. (1 pt)

Geothermal heat. (1 pt)

[TB 5] 11) The water temperature in Lake Vostok is about -3 deg C. Provide one possible explanation for the lowered freezing point. (*Hint: consider the relative densities of water and ice*) (3 pts)

Ice is less dense than water. Thus the melting point of ice decreases as pressure increases (negative slope on P-T phase diagram).

Lake Vostok is under very high pressure from the thick ice sheet above it. This depresses the freezing point of water. (3 pts)

OR 1.5 pts for the explanation that Lake Vostok is hypersaline, which also depresses the freezing point.

Let us consider the volume budget of Lake Vostok.

12) Assuming that the surrounding bedrock is impermeable, what is the only source of water for Lake Vostok? (1 pt)

The ice above. (1 pt)

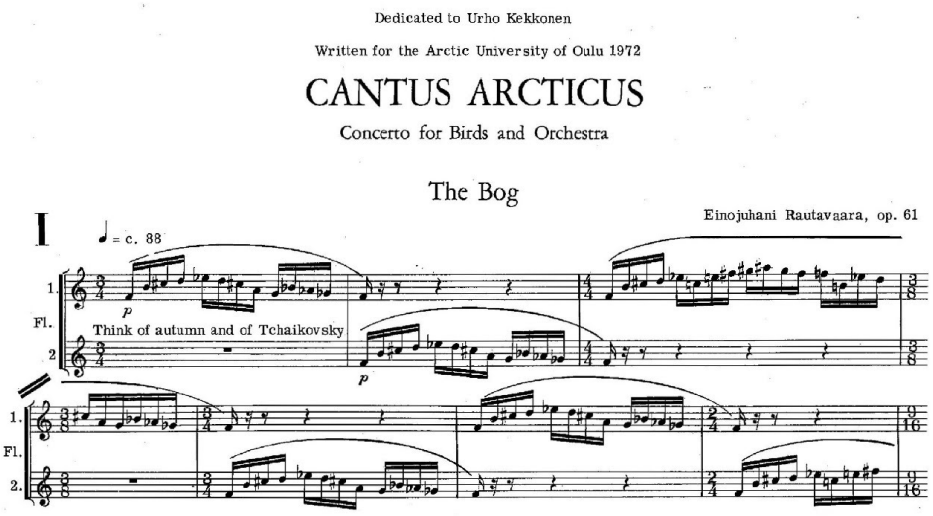
13) Considering your answer in (12), describe what condition must hold for the volume of Lake Vostok to remain constant. *(Hint: consider dynamic equilibrium and the fact that there is an external heat source)* (2 pts)

Since there is geothermal heat warming up the bottom water, there must be an equal rate of ice formation and ice loss at the water-ice interface. (2 pts)

14) Information about ancient lakes can be gleaned from their deposited sediment through analysis of proxies. What is a proxy? Provide one example. (2 pts)

A proxy is some physical, chemical, or biological signature that corresponds with changes in a physical/chemical parameter (1 pt). Examples of proxies include pollen grains, oxygen isotopes, carbonate isotope clumping, etc (1 pt).

Section 4 (10 pts)



Rautavaara was a composer from Finland, which has a lot of bogs. Answer the following questions about bogs. (1 pt each)

1) **True**/false: bogs act as carbon sinks

2) True/**false**: bogs are typically more alkaline, while fens are typically more acidic

3) **True**/false: the formation of peat in bogs is aided by anoxic conditions

4) True/**false**: bogs typically form in regions of high groundwater permeability

5) The Hanford Site in Washington was chosen in part due to its proximity to the Columbia River. Briefly describe two ways in which the Hanford Site may pollute the Columbia River. (2 pts)

Leaching of radioactive isotopes (Tc-99, etc.) into the river (1 pt) and thermal pollution from cooling the reactor (1 pt).

6) Briefly explain the effect of river damming on upstream sedimentation. Include base level in your explanation. (2 pts)

Damming a river creates a new local base level (1 pt), which prompts the deposition of sediment in the dam reservoir (1 pt).

7) Briefly describe how a natural levee is formed. (2 pts)

Repeated cycles of flooding bring sediment to line the pre-flood riverbank, building a levee. (2 pts)